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RESEARCH ARTICLE

Enhancing the Mechanical Properties of Solid Electrolytes for Crack-Resistant Cathodes in All-Solid-State Batteries

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ABSTRACT

The electrochemical performance of all-solid-state batteries (ASSBs) is strongly influenced by the physicochemical characteristics of solid electrolytes; however, the impact of their mechanical properties on cell performance remains comparatively underexplored. Here, SiO_2 incorporation during Li_6PS_5Cl synthesis induces Si/O co-substitution, forming $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$ with modified mechanical properties. Combined theoretical and experimental investigations show an increase in Young's modulus from 16 to 21 GPa upon Si/O co-substitution. AFM nanoindentation analysis further reveals higher elastic stiffness and hardness for $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$. Cells employing $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$-based composite cathodes exhibit markedly improved cycling performance, achieving a capacity retention of 90.8% after 500 cycles, surpassing that of cells based on conventional Li_6PS_5Cl. Overall, this study underscores the importance of structural engineering for tailoring the mechanical properties of solid electrolytes and provides a design framework for developing mechanically robust composite cathodes for next-generation ASSBs.

1 | Introduction

All-solid-state batteries (ASSBs) have emerged as promising energy storage systems owing to their superior safety and high energy density [1, 2]. Sulfide-based solid electrolytes (SEs), represented by Li_6PS_5Cl (LPSCl), offer the combined advantages of excellent ionic conductivity (exceeding 1.0 mS cm^{-1}) and mechanical ductility. These favorable properties enable intimate contact at the cathode-electrolyte interfaces under fabrication pressure, which facilitates efficient Li^+ transport [3–5]. In contrast, the lattice of the layered cathode active material, $LiNi_{0.7}Co_{0.15}Mn_{0.15}O_2$ (NCM), undergoes a volume contraction of ~8% while charging. This contraction induces internal mechanical stress and leads to interfacial delamination, thereby degrading the electrochemical performance [6]. Several strategies, such as the application of high external pressure [3, 7, 8], and the use of submicron solid electrolyte particles [9–11] and high-adhesion binders [12–14], have been proposed to preserve intimate solid-solid interfacial contact. Nevertheless, cycling-induced mechanical stress remains a critical bottleneck because it limits the long-term stability and retention of low interfacial resistance at the electrode level.

The mechanical properties of solid electrolytes thus play a pivotal role in determining the electrochemical performance of ASSBs. This deformable nature enables sulfide solid electrolytes to undergo pressure-induced densification, thereby improving

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interfacial contact [15, 16]. A low Young's modulus promotes cathode densification during cell fabrication but renders the electrolyte susceptible to crack initiation and structural degradation during cycling-induced mechanical stress under pressurized conditions [17–19]. These observations suggest that an optimal balance between deformability and mechanical robustness is required for stable electrode operation. Therefore, a moderate increase in the modulus and elasticity of a solid electrolyte can mitigate irreversible deformation and preserve electrode structural stability during repeated cycling, thereby maintaining stable cathode–electrolyte interfacial contact [19, 20]. However, the excessively high moduli of oxide-based solid electrolytes impede sufficient deformation, which hinders the formation of intimate interfaces necessary for efficient Li^+ transport [21–24]. Previous studies therefore emphasized the critical need to optimize the modulus of solid electrolytes, particularly for SEs integrated with high-capacity cathodes that undergo large volume changes [25, 26]. Based on these observations, an enhanced Young's modulus would enable the solid electrolyte to elastically accommodate cathode-induced volume changes without irreversible deformation, thereby mitigating stress accumulation and preserving interfacial contact during cycling. This suggests that tailoring the modulus of SEs is essential to ensure long-term cycling stability and an extended cycle life.

In this study, a robust solid electrolyte with enhanced mechanical properties was synthesized by incorporating amorphous SiO_2 into the solid electrolyte Li_6PS_5Cl (LPSCl) via a mixing-annealing process. The incorporation of SiO_2 resulted in the formation of $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$ (LPSiSOCl), in which the P and S atoms in the PS_4^{3-} polyhedron are selectively substituted by Si and O atoms from SiO_2, respectively. The positive effects of the co-substitution by Si and O on the mechanical and electrochemical properties were comprehensively investigated via theoretical predictions and experimental validation. Based on the various structural and electrochemical analyses, we demonstrated that compositional engineering provides a viable pathway to enhance interfacial stability and extend the operational durability of ASSBs.

2 | Results and Discussion

Solid electrolyte (SE) powder samples with the general formula $Li_{6+x}P_{1-x}Si_xS_{5-2x}O_{2x}Cl$ ($x = 0$, 0.1, 0.2, and 0.3) were synthesized by heating a mixture of Li_2S, P_2S_5, LiCl, and amorphous SiO_2. This served to introduce Si^{4+} and O^{2-} into the argyrodite framework of LPSCl during the synthesis of LPSiSOCl, with amorphous SiO_2 acting as the source of the dopants. The $Li_{6+x}P_{1-x}Si_xS_{5-2x}O_{2x}Cl$ powder was prepared as follows (see Figure S1): (i) Li_2S, P_2S_5, LiCl, and SiO_2 powders were mixed using planetary ball milling, (ii) the resulting mixture was calcined at 550°C for 2 h under vacuum, and (iii) particle-size control was exercised under optimized conditions to yield the $Li_{6+x}P_{1-x}Si_xS_{5-2x}O_{2x}Cl$ powder. As illustrated in Figure 1a, Si^{4+} preferentially substituted P^{5+} at the 4b sites, whereas O^{2-} occupied the S^{2-} positions at the 16e sites within the PS_4^{3-} tetrahedrons [27–29]. X-ray diffraction (XRD) analysis confirmed the formation of a phase-pure argyrodite structure for $Li_{6+x}P_{1-x}Si_xS_{5-2x}O_{2x}Cl$ (x = 0, 0.1, 0.2) with no evidence of secondary phases (Figure 1b). However, at a SiO_2 ratio of x = 0.3, the argyrodite phase was identified together with additional peaks corresponding to Li_3PO_4 (22.3° and 23.1°) and LiCl (34.9°), indicating that the substitution exceeded the solubility limit of the argyrodite structure and resulted in the formation of impurity phases [30, 31].

The ionic conductivities, measured by AC impedance (Figure 1c), varied according to the composition. Specifically, the conductivities of the series $Li_{6+x}P_{1-x}Si_xS_{5-2x}O_{2x}Cl$ (x = 0.1, 0.2, and 0.3), which were determined to be 1.23, 1.17, and 0.66 mS cm^{-1}, respectively, were slightly lower than that of Li_6PS_5Cl (Table S1). Among these compositions, the $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$ (x = 0.1) powder exhibited moderate conductivity without detectable insulating secondary phases due to the optimized level of Si and O substitution. Additionally, we confirmed that SiO_2 incorporation is also compatible with halogen-rich argyrodite compositions. Specifically, halogen-rich $Li_{5.8}P_{0.9}Si_{0.1}S_{4.5}O_{0.2}Cl_{1.3}$ was successfully synthesized, exhibiting an ionic conductivity of 3.8 mS cm^{-1} at room temperature, as shown in Figure S2. This result suggests that further compositional optimization could enable simultaneous enhancement of mechanical robustness and ionic conductivity through SiO_2 incorporation.

The incorporation of O into the PS_4^{3-} group reportedly lowers the ionic conductivity because the stronger P−O bonds increase the interaction between Li^+ and the polyhedral framework, thereby increasing the Li^+ migration barrier and lowering the Li^+ conductivity. At the same time, it also enhances the structural stability against H_2O reactivity because O^{2-} is more electronegative than S^{2-} [32–34]. In contrast, the substitution of P^{5+} by Si^{4+} increases the Li^+ concentration and promotes S^{2-}/Cl^- disorder within the lattice to enhance the Li^+ conductivity [35, 36]. Thus, the combined incorporation of Si and O is beneficial, with these two species complementing each other to improve the properties of the solid electrolyte. Among the doped electrolytes, $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$ (x = 0.1), which exhibited the highest ionic conductivity and was devoid of detectable impurities, was selected as the optimal composition, and its particle morphology and the chemical states of Si and O were subsequently analyzed.

Scanning electron microscopy (SEM) images of the solid electrolyte powders Li_6PS_5Cl and $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$ (hereafter denoted as LPSCl and LPSiSOCl) and particle size analyses revealed average particle sizes of 1.60 and 1.62 μm (D_{50}), respectively (Figure 1d–f and Figures S3 and S4). This indicated that the inclusion of SiO_2 did not affect the morphology of LPSiSOCl. The chemical states of Si and O in LPSiSOCl were determined by X-ray photoelectron spectroscopy (XPS) (Figure 1g,h). The P 2p spectrum of LPSCl exhibited the characteristic peaks of PS_4^{3-} at 132.2 and 133.0 eV [37–39]. In contrast, LPSiSOCl had an additional peak at 134.2 eV, which corresponds to PS_3O^{3-}, along with the PS_4^{3-} peaks observed for LPSCl [40, 41]. Distinct peaks assigned to the Si−O (101.6 eV) and Si−S (102.4 eV) bonds were observed for LPSiSOCl [42, 43]. By comparison, the signal of amorphous SiO_2 typically occurs near 104.5 eV [44]. Additionally, Figure S5 shows that co-substitution by Si and O did not affect the chemical states of S and Li as evidenced by the XPS Li 1s and S 2p spectra.

Solid-state ^{31}P magic angle spinning nuclear magnetic resonance (MAS NMR) spectroscopy enabled the local chemical environments and structural changes induced by co-substitution with Si and O to be investigated (Figure 1i). The ^{31}P MAS NMR spectrum

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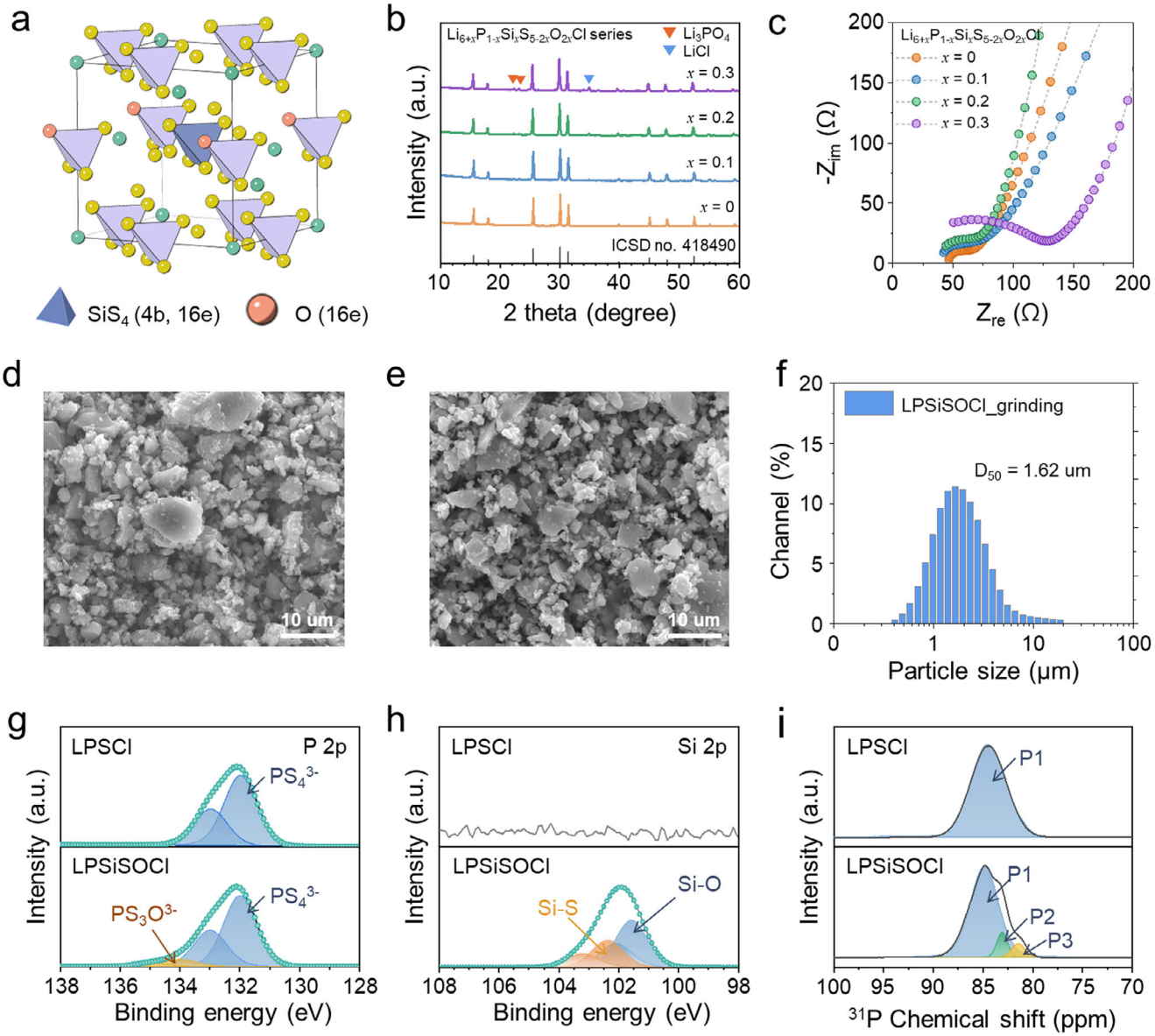


FIGURE 1 | Structural analysis of solid electrolyte $Li_{6+x}P_{1-x}Si_xS_{5-2x}O_{2x}Cl$. (a) Schematic illustration of the structure of $Li_{6+x}P_{1-x}Si_xS_{5-2x}O_{2x}Cl$, (b) X-ray diffraction (XRD) patterns and (c) Nyquist plots of $Li_{6+x}P_{1-x}Si_xS_{5-2x}O_{2x}Cl$ powders (x = 0, 0.1, 0.2, and 0.3), SEM images of (d) Li_6PS_5Cl (LPSCl) and (e) $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$ (LPSiSOCl) powders, (f) particle size distribution of LPSiSOCl powder, X-ray photoelectron spectroscopy (XPS) (g) P 2p and (h) Si 2p spectra, and (i) ^{31}P magic angle spinning nuclear magnetic resonance (MAS NMR) spectra of LPSCl and LPSiSOCl powders.

of LPSiSOCl exhibited P1, P2, and P3 resonances consistent with an increased degree of S^{2-}/Cl^- disorder around the P centers in the PS_4 units. These results were in agreement with previous observations that O substitution typically alters the local chemical environment of phosphorus [45–47], and confirmed that Si and O were successfully incorporated into the argyrodite structure as a result of the thermal treatment.

The mechanical properties of solid electrolytes are critically linked to the volume expansion and contraction of the cathode active materials during cycling. For NCM, the lattice contracts and expands by approximately 8 vol% during cycling, thus transmitting mechanical stress from the NCM to the surrounding LPSCl. With repeated cycling, these stresses accumulate locally to induce the formation of microcracks within the electrodes and eventual performance degradation. Alleviating the localized stress originating from NCM during cycling was therefore necessary; thus, increasing the Young's modulus of the SE seemed an attractive strategy.

Figure 2a presents the density functional theory (DFT) predictions of the bulk moduli (B_0) of various argyrodite frameworks (Figure 2a, upper rows) doped with Si and O, and demonstrates the mechanical reinforcement imparted by these dopants. The bulk modulus was calculated by fitting the energy–volume (E–V) data derived from DFT calculations to the Murnaghan equation of state. As shown in Figure 2a, both single-element substitution (either Si or O) and co-substitution by Si and O resulted in bulk moduli higher than that of Li_6PS_5Cl. Specifically, relative to Li_6PS_5Cl (23.66 GPa), the co-substituted $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$ (LPSiSOCl) exhibited an increased bulk modulus of 26.7 GPa. Consistently, the Si-only and O-only substituted structures

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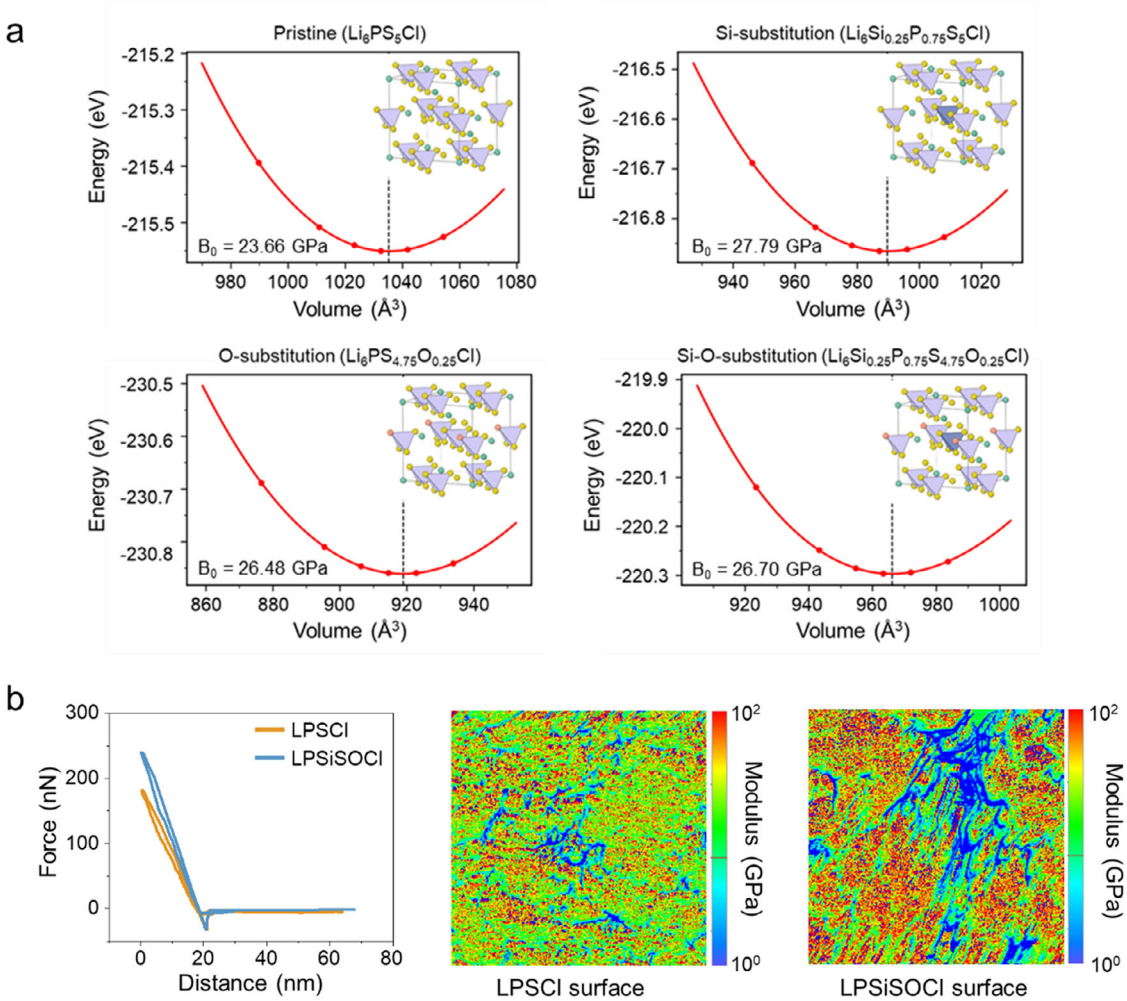


FIGURE 2 | Mechanical properties of solid electrolyte $Li_{6+x}P_{1-x}Si_xS_{5-2x}O_{2x}Cl$. (a) Density functional theory-derived bulk modulus values for the Li_6PS_5Cl, $Li_6Si_{0.25}P_{0.75}S_5Cl$, $Li_6PS_{4.75}O_{0.25}Cl$, and $Li_6Si_{0.25}P_{0.75}S_{4.75}O_{0.25}Cl$ structures. (b) Young's modulus mapping results for Li_6PS_5Cl (LPSCl) and $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$ (LPSiSOCl).

yielded bulk moduli of 27.79 and 26.48 GPa, respectively, both of which are higher with respect to that of Li_6PS_5Cl. These computational results clearly indicate that co-substitution by Si and O effectively reinforces the mechanical modulus of the argyrodite framework. Building on these computational insights, we subsequently determined whether the substitution-induced mechanical enhancement was reflected in the effectiveness of the interfacial contact.

To validate the computational predictions, the Young's moduli of Li_6PS_5Cl (LPSCl) and $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$ (LPSiSOCl) were experimentally measured using atomic force microscopy (AFM) operated in PeakForce quantitative nanomechanical mapping (QNM) mode. QNM provided spatially resolved maps of the local Young's modulus, and the corresponding histograms quantified the distribution thereof across the sample surfaces. The AFM measurements revealed a higher Young's modulus of 21.2 GPa for LPSiSOCl, compared with 16.6 GPa for LPSCl (Figure 2b). This notable increase in Young's modulus could be attributed to the incorporation of stronger Si–S, Si–O, and P–O bonds within the framework, which likely reinforce the mechanical stiffness of the argyrodite framework [48–50]. In addition, AFM nanoindentation P–h curves, as shown in Figure S6, revealed distinct elastic recovery behaviors between LPSCl and LPSiSOCl. Specifically, LPSCl exhibited a larger residual depth (46.4 nm) than LPSiSOCl (35.7 nm), indicating an approximately 10.7 nm (≈23%) reduction in irreversible deformation for LPSiSOCl. The smaller hysteresis observed for LPSiSOCl further indicates enhanced elastic recovery behavior [51, 52]. To further quantify the mechanical properties, the diamond-tip nanoindentation curves were analyzed using the Oliver–Pharr method (see Figure S6). The Young's moduli of LPSCl and LPSiSOCl were determined to be 28.6 and 37.6 GPa, respectively, assuming a Poisson's ratio of 0.30, reproducing the same relative trend observed by PeakForce QNM. More importantly, the corresponding hardness values were 1.34 and 1.56 GPa for LPSCl and LPSiSOCl, respectively. The higher hardness of LPSiSOCl indicates improved resistance to indentation-induced irreversible deformation, consistent with its smaller residual indentation depth. Collectively, these results demonstrate the distinct mechanical characteristics of LPSiSOCl, including increased elastic stiffness and resistance to irreversible deformation.

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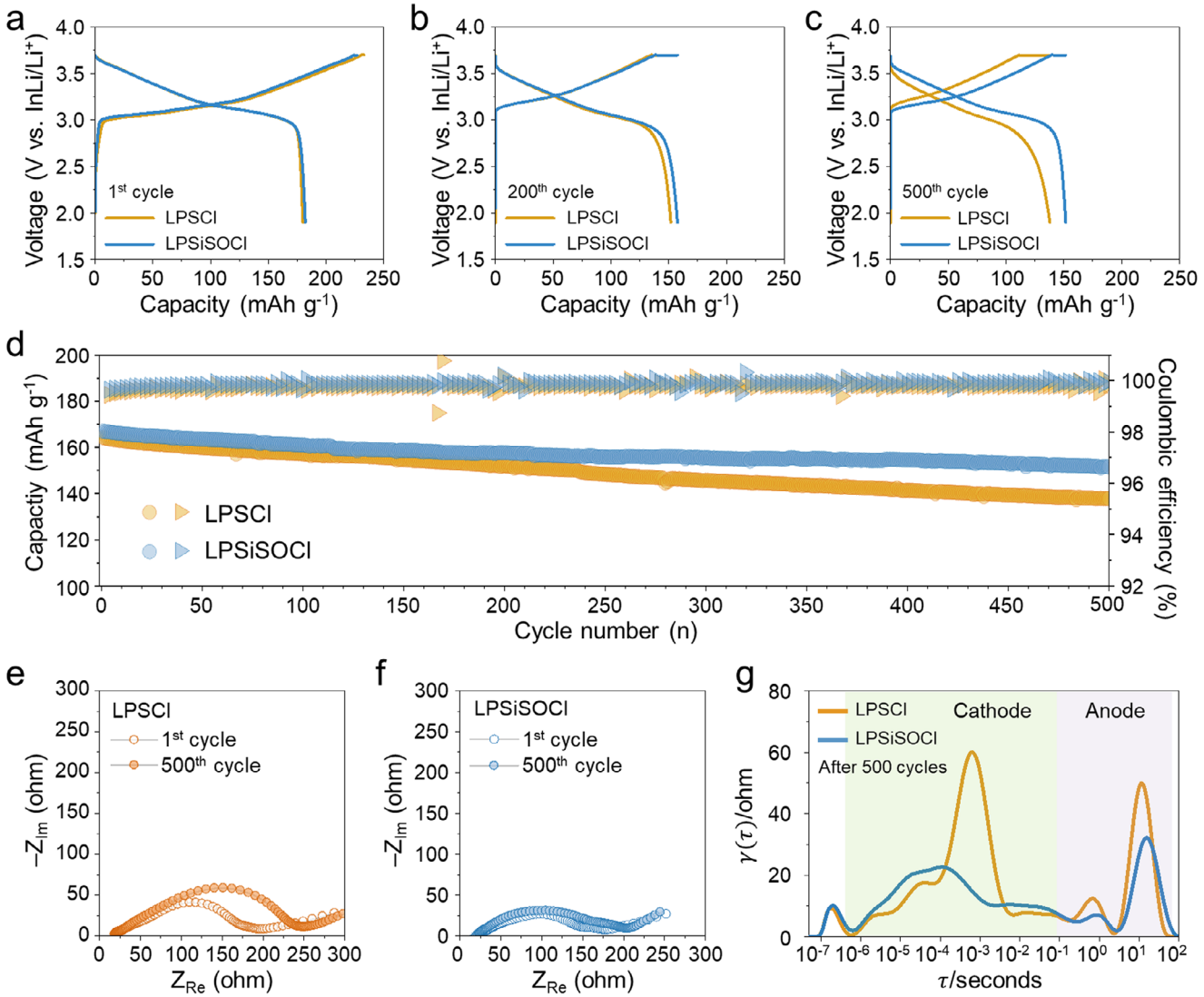


FIGURE 3 | Electrochemical performance of the cathodes with LPSCl and LPSiSOCl. Voltage profiles of the (a) 1st, (b) 200th, and (c) 500th cycles. (d) Cycling performance from the 2nd to the 500th cycles. Electrochemical impedance spectroscopy (EIS) measurements: (e,f) Nyquist plots with cathodes containing LPSCl and LPSiSOCl after the 1st and 500th cycles, and (g) distribution of relaxation time (DRT) transformed from Nyquist plots after the 500th cycle.

To assess the electrochemical performance associated with the Young's moduli of the SEs, two composite cathodes containing LPSCl or LPSiSOCl were prepared and evaluated using powder-pressed cells with Li—In alloy anodes. Figure 3a–d compares the galvanostatic voltage profiles at selected cycles and the cycling performance for 500 cycles. The first cycle was performed at 0.1 C for formation, and this was followed by 500 cycles at 0.5 C. The electrochemical performance of both of the composite cathodes was similar in the first cycle. The initial charge and discharge capacities for the cathode with LPSCl were 232.8 and 180.0 mAh g^{-1}, respectively, whereas those for the composite cathode with LPSiSOCl were 227.0 and 182.1 mAh g^{-1}, respectively. At the 200th cycle, both composite cathodes still maintained similar electrochemical behaviors: the discharge capacities for the cathodes with LPSCl and LPSiCOCl were 151.8 and 157.5 mAh g^{-1}, respectively. After 500 cycles, the electrochemical performance of the cathode with LPSiSOCl remained stable with a discharge capacity of 151.6 mAh g^{-1} but the composite cathode with LPSCl experienced noticeable capacity loss (137.8 mAh g^{-1}) along with a clear increase in the average charge–discharge overpotential.

The assessment of the cycling performance of the composite cathodes with LPSCl and LPSiSOCl at 0.5C for 500 cycles (Figure 3d) attested to the superior cycling stability of the composite cathode with LPSiSOCl. After 500 cycles, the capacity retention of the composite cathodes with LPSCl and LPSiSOCl was 83.9% and 90.8%, and their average Coulombic efficiencies over 500 cycles were 99.8% and 99.9%, respectively. As indicated by the cycle profiles, the difference in resistance between the two composite cathodes progressively increased with cycling. As shown in Figure S7, the cathode with LPSiSOCl consistently exhibited improved cycling stability compared with that containing pristine LPSCl even at high current densities of 1 and 2 C. After 100 cycles at 1 C, the cathode with LPSCl retained a discharge capacity of 128.1 mAh g^{-1}, whereas the cathode with LPSiSOCl delivered a higher capacity of 148.2 mAh g^{-1}. Similarly, at 2 C, the discharge

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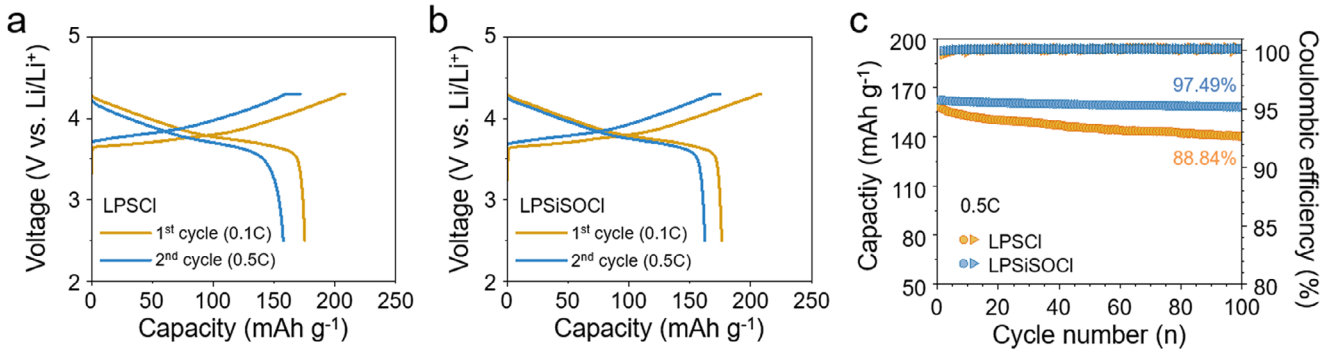


FIGURE 4 | Electrochemical performance of the cathodes under a low stack pressure of 7 MPa. Voltage profiles of the cathodes with (a) LPSCl and (b) LPSiSOCl, and (c) cycling performance for 100 cycles.

capacities after 100 cycles were 112.8 and 136.2 mAh g^{-1} for LPSCl and LPSiSOCl, respectively. Specifically, after 100 cycles at 3 C, the LPSCl- and LPSiSOCl-based cathodes retained 75.3% and 91.3% of their initial capacities, respectively, while the corresponding capacity retentions at 5 C were 64.5% and 85.5%. At higher C-rates, the more rapid delithiation of the NCM particles can induce faster and more spatially heterogeneous lattice-volume changes, thereby promoting interfacial contact degradation. In addition, the differential capacity (dQ/dV) curves derived from the cycling profiles revealed that the overall resistance and capacity fading of the cathode progressively increased for LPSCl (Figure S8) [53–55]. In contrast, the rate capability tests indicated no significant performance difference between the two composite cathodes over the range from 0.1 to 2.0 C, which is attributable to the similar morphologies and ionic conductivities of LPSCl and LPSiSOCl (Figure S9).

Electrochemical impedance spectroscopy (EIS) was used to measure the internal resistance of both composite cathodes to pinpoint the origin of the performance differences between the LPSCl- and LPSiSOCl-based composite cathodes. The Nyquist plots in Figure 3e,f show that the total impedance of the composite cathode with LPSCl was noticeably higher after 500 cycles compared to that after the first cycle, which implies an increase in the overall resistance of the composite cathode with LPSCl. In contrast, the total impedance of the composite cathode with LPSiSOCl remained largely unchanged, even after 500 cycles. The distribution of relaxation time (DRT) method was used to deconvolute the various impedance contributions to the Nyquist plots (Figure 3g). Depending on the relaxation time (τ), the DRT data could be divided into regions corresponding to the cathode and anode. At the cathode, the responses at 10^1–10^4 Hz were attributed to the cathode-electrolyte interface [56–59]. Consistent with the EIS results, the DRT analysis conducted after 500 cycles revealed an intense peak associated with the cathode-electrolyte interface for the cathode with LPSCl. This suggested that the higher charge-transfer resistance (R_{ct}) measured for the composite cathode with LPSCl compared to that of the composite cathode with LPSiSOCl could be ascribed to the cathode-electrolyte interface.

Additionally, even under the reduced stack pressure of 7 MPa, the cathode with LPSiSOCl still demonstrated superior cycling performance compared with the cathode containing LPSCl, consistent with the trend observed under high-pressure conditions (see Figure 4). The cathodes with LPSCl and LPSiSOCl delivered initial discharge capacities of 175.5 and 176.1 mAh g^{-1} at 0.1 C during the formation cycle, respectively, and retained 88.8% and 97.5% of their initial capacities after 100 cycles at 0.5 C. These results suggest that mechanically reinforced solid electrolytes can more effectively tolerate internally generated stress arising from repeated volume changes of cathode materials, thereby preserving stable interfacial contact and structural integrity within composite electrodes even under reduced stack-pressure conditions.

To further distinguish interfacial electrochemical degradation from mechanically induced degradation, we performed a control experiment with the upper cutoff voltage limited to 3.8 V (vs. Li/Li^+). In this voltage range, pronounced NCM contraction is largely suppressed, while oxidative decomposition of the sulfide solid electrolyte and associated CEI formation can still occur [26, 60, 61]. Under a stack pressure of 7 MPa, the LPSCl- and LPSiSOCl-based cathodes exhibited nearly identical voltage profiles and stable cycling with negligible capacity decay over 50 cycles (Figure S10). In contrast, a pronounced difference in cycling stability emerged upon charging to 4.3 V (Figure 4), at which substantial NCM contraction occurs. This comparison is consistent with a significant contribution of the mechanical characteristics of LPSiSOCl to the improved cycling stability under deep-charge conditions.

To further examine the effect of SiO_2 incorporation on the structural stability of the sulfide solid electrolyte, the synthesized electrolytes were exposed to an atmosphere with 50% relative humidity for 1 h, followed by measurements of H_2S generation and ionic conductivity. Compared to pristine LPSCl, LPSiSOCl exhibited lower H_2S generation (76.3 ppm vs. 120.8 ppm) and retained a higher ionic conductivity after exposure (0.57 vs. 0.13 mS cm^{-1}), indicating enhanced air stability upon SiO_2 incorporation (see Figure S11). In addition, composite cathodes prepared using the moisture-exposed electrolytes showed that the LPSiSOCl-based cathode delivered superior rate capability compared to the pristine LPSCl-based cathode, as shown in Figure S12. These results demonstrate that SiO_2 incorporation improves not only the moisture stability of sulfide solid electrolytes, but also their long-term electrochemical performance.

Cross-sectional SEM-EDS analyses were conducted to elucidate the origin of the differences in electrochemical performance between the two composite cathodes before cycling and after 500

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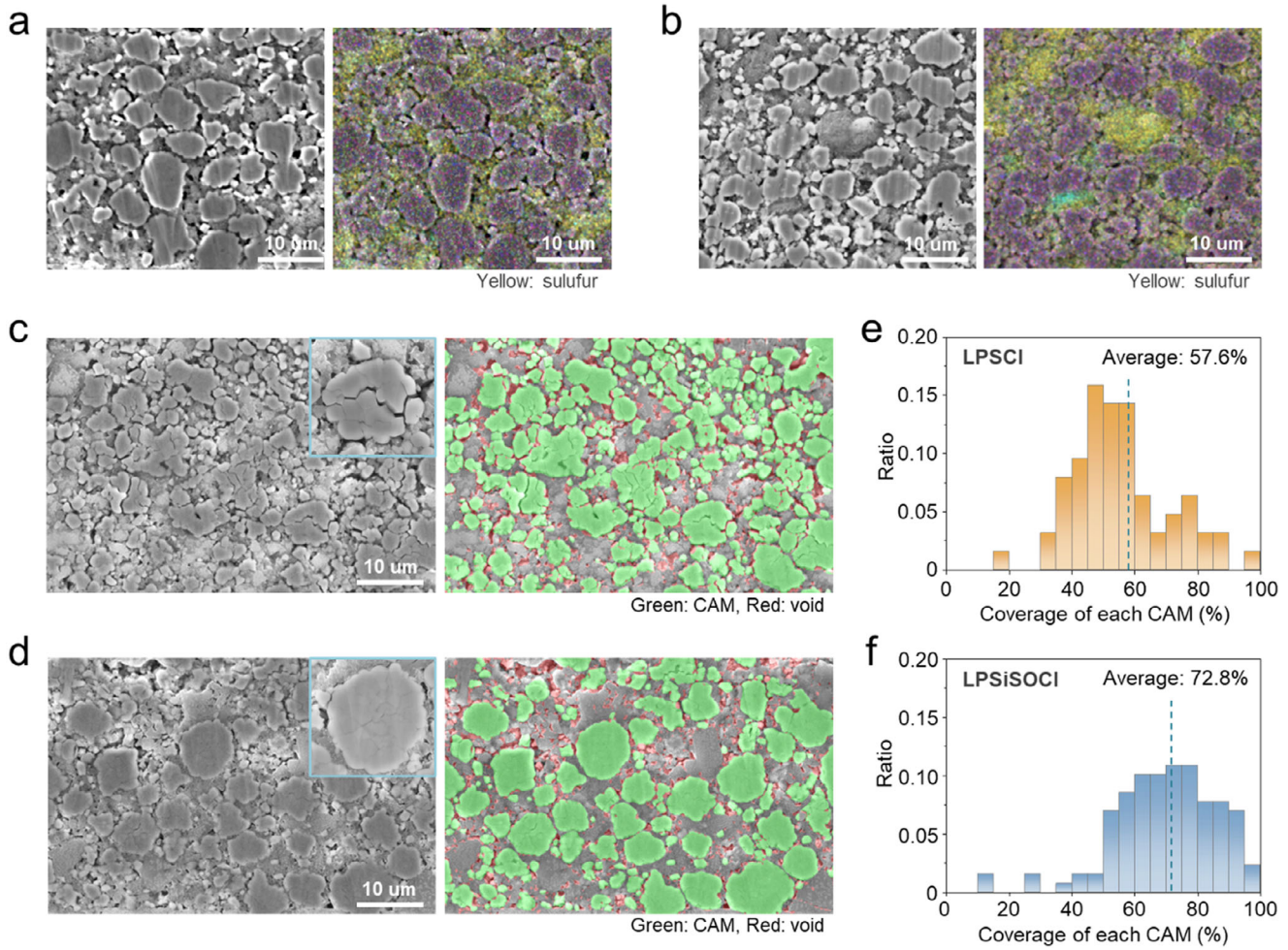


FIGURE 5 | Ex situ SEM-EDS analysis of composite cathodes. Cross-sectional SEM images and EDS elemental maps of composite cathodes with (a) LPSCl and (b) LPSiSOCl before cycling, and after 500 cycles for (c) LPSCl and (d) LPSiSOCl. (e,f) Quantitative evaluation of cathode material coverage obtained from (c) and (d).

cycles. Cross-sectional SEM-EDS images (Figure 5a,b) confirmed comparable cathode–solid electrolyte interfacial contact in both cathodes before cycling. After 500 cycles, the cathode with LPSCl experienced severe microcracking throughout the composite cathode and within the NCM particles, whereas the morphology of the cathode with LPSiSOCl remained relatively intact. This morphological difference was also clearly evidenced by the EDS elemental mapping results (Figure S13).

Consistent with previous studies showing that the cycling performance of composite cathodes correlates with the contact area between the cathode active materials and solid electrolytes, we applied a previously developed image analysis technique to quantify this coverage in cycled LPSCl and LPSiSOCl from the SEM images [62–64]. Using this technique, the contrast and brightness of the cross-sectional SEM images were carefully adjusted. NCM particles and void spaces were colored green and red, respectively, for clear differentiation. Then, the perimeters of regions covered by solid electrolyte and those occupied by voids were measured to calculate the average coverage of each composite cathode. Prior to cycling, quantitative coverage analysis based on cross-sectional SEM-EDS images (Figure 5a,b) confirmed comparable cathode–solid electrolyte interfacial contact in both electrodes, with coverage values of 76.8% and 75.7% for the cathodes containing LPSCl and LPSiSOCl, respectively (Figure S14). The processed images after 500 cycles were digitally segmented to distinguish the regions containing active material from regions occupied by voids and solid electrolyte, as shown in Figure 5c,d. The average coverage values for the cathodes with LPSCl and LPSiSOCl were determined to be 57.6% and 72.8%, respectively (Figure 5e,f). The cathode containing LPSiSOCl therefore maintained a larger contact area between the cathode material and solid electrolyte after prolonged cycling.

This improved contact retention is consistent with the mechanical characteristics of LPSiSOCl. Repeated NCM volume changes, particularly the pronounced contraction at high states of charge, can generate local chemo-mechanical stresses and lead to interfacial contact loss within the composite cathode. Under cycling-induced chemo-mechanical stress, its higher Young's modulus and hardness reflect higher elastic stiffness and increased resistance to irreversible deformation, respectively. These mechanical characteristics are consistent with the retention of interfacial contact between NCM and the solid electrolyte observed after prolonged cycling, which may in turn support continuous Li-ion transport pathways within the composite cathode. In practice,

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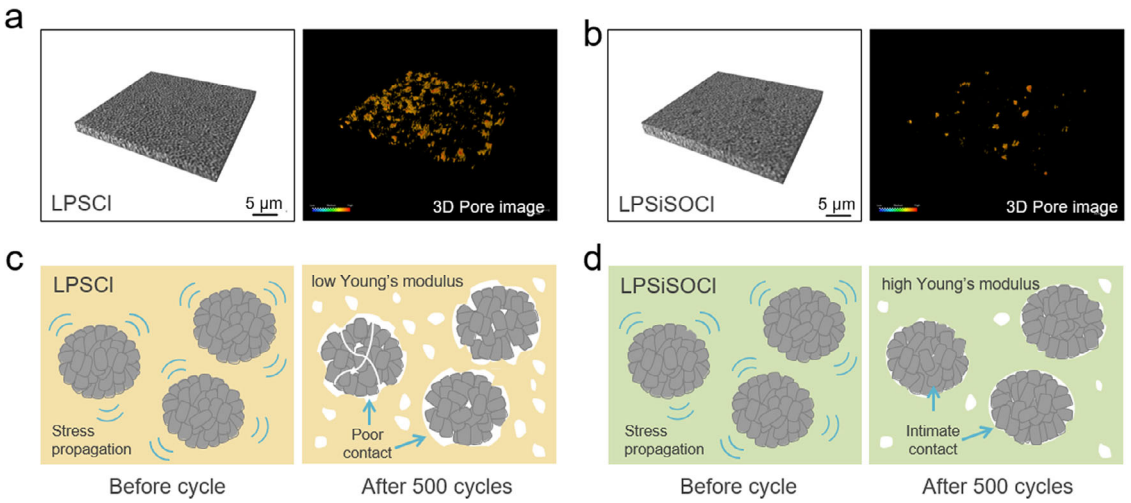


FIGURE 6 | Visualization of pore evolution and proposed cycling mechanism. 3D XRM images and corresponding pore-segmented reconstructions of the cathodes with (a) LPSCl and (b) LPSiSOCl after 50 cycles. (c,d) Schematic illustrations of the proposed long-term cycling mechanisms in solid electrolytes with improved mechanical properties.

the electrochemical performance of ASSB cathodes is governed not only by the physical properties of the composite cathode but also by chemical degradation at the cathode-electrolyte interface. To purposefully minimize the effect of chemical degradation, the surfaces of NCM particles were coated with a layer of $LiNbO_3$ [65, 66]. After 50 cycles, the XPS spectra of both composite cathodes were nearly identical, indicating that unfavorable interfacial reactions were negligible in this work (Figure S15). Consequently, the performance differences observed in the long-term cycling tests could be primarily attributed to mechanical degradation within the composite cathode arising from the mechanical properties of the solid electrolytes.

To further investigate the internal structural evolution of the composite electrodes during cycling, 3D nano-XCT analysis was additionally performed after 50 cycles. Unlike conventional SEM analysis, which is limited to two-dimensional surface characterization, nano-XCT enables direct visualization and quantification of internal pore formation within the electrode structure. As presented in Figure 6a,b, quantitative analysis of sub-micrometer-scale pores revealed that the electrode using the SiO_2-containing solid electrolyte exhibited an internal porosity of ~8.7%, whereas the electrode employing pristine LPSCl showed a significantly higher porosity of ~16.2%. These results indicate that the mechanically reinforced LPSiSOCl effectively suppresses internal void formation and helps maintain the structural integrity of the composite electrode during repeated cycling.

Figure 6c,d schematically illustrate the effect of the mechanical properties of the SEs on their post-cycling working mechanisms. Before cycling commences, the interfacial contact areas between the cathode active material and the solid electrolyte are essentially the same for both SEs because the size of this area strongly depends on the particle size and composition ratio. During repeated cycling, however, the cathode active material undergoes continuous expansion and contraction, which generates mechanical stress. Under such cycling-induced mechanical loading, both the elastic stiffness and resistance to irreversible deformation of the SEs become important [20, 67]. Therefore, Young's modulus should be considered together with other mechanical properties and interfacial compliance. LPSiSOCl exhibits a higher Young's modulus and hardness than LPSCl, indicating higher elastic stiffness and improved resistance to indentation-induced irreversible deformation, respectively. These properties help limit mechanical deformation during repeated cathode volume changes and are consistent with the reduced mechanical degradation and larger cathode–solid electrolyte contact area observed after prolonged cycling. The improved structural stability and contact retention contribute to its enhanced long-term cycling performance.

3 | Conclusions

In this study, we developed the solid electrolyte $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$ by incorporating SiO_2 as a dopant when synthesizing the electrolyte, thus enabling the mechanical properties to be markedly enhanced. DFT calculations combined with AFM measurements elucidated that co-substitution by Si and O substantially increases the bulk modulus relative to that of Li_6PS_5Cl. Nanoindentation analysis further revealed increased hardness, indicating improved resistance to irreversible deformation. These results firmly establish compositional engineering as an effective route to reinforce the mechanical integrity of argyrodite solid electrolytes. Importantly, the mechanically strengthened $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$ contributed to sustaining intimate interfacial contact with the cathode material. The enhanced mechanical properties of $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$ helped maintain the composite-cathode microstructure and interfacial contact during repeated volume fluctuations of the cathode active material during extended cycling. Consequently, the long-term cycling stability of the composite cathode containing this electrolyte was substantially superior to that of the Li_6PS_5Cl-based composite cathode. These findings highlight that precise control of the mechanical properties of solid electrolytes is a powerful strategy for improving the performance of ASSBs and provide a promising design principle for developing robust solid electrolytes that combine high mechanical robustness with excellent electrochemical functionality.

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4 | Experimental Section

4.1 | Material Preparation

Argyrodite-type Li_6PS_5Cl (LPSCl) and $Li_{6.1}P_{0.9}Si_{0.1}S_{4.8}O_{0.2}Cl$ (LPSiSOCl) were prepared by a mechanochemical route followed by heat treatment. In a dry room, stoichiometric amounts of Li_2S (99.9%, Alfa Aesar), P_2S_5 (99%, Sigma-Aldrich), LiCl (99%, Sigma-Aldrich), and amorphous SiO_2 (99.9%, Alfa Aesar; for LPSiSOCl) were weighed and premixed. The blend was transferred to a ZrO_2 jar and milled in a planetary ball mill (Pulverisette 6, Fritsch) at 300 rpm for 2 h. The resulting powder (3 g) was uniaxially pressed at 50 MPa and annealed at 550°C for 6 h under vacuum to yield argyrodite LPSCl (the same protocol was used for LPSiSOCl). Finally, the pellets were ground by blade milling (SM-3L, Hsiang Tai) at 28000 rpm for ten 30 s cycles interspersed with 1-min rest periods, followed by ball milling with ZrO_2 balls at 450 rpm for 1 h.

4.2 | Structural Characterizations

Scanning electron microscopy (SEM, JSM-IT200, JEOL) and a particle size analyzer (PSA, S3500 series, MICROTRAC) were used to characterize the morphology and particle size distribution of LPSCl and LPSiSOCl, respectively. For PSA measurements, the prepared powders were dispersed in xylene with Tween 80 as a dispersant, followed by sonication to ensure uniform dispersion prior to analysis. Powder X-ray diffraction (XRD, Empyrean, PANalytical) was used to identify the crystalline phases of LPSCl and LPSiSOCl. X-ray photoelectron spectroscopy (XPS, K-ALPHA, Thermo Fisher Scientific) enabled their chemical states to be analyzed. Cross sections of the cathode composites were prepared for observation by polishing them with a cryo-cross-sectional polisher (IB-19520CCP, JEOL) and then observed by SEM (JSM-IT200, JEOL) in a dry room at −50°C. Modulus mapping measurements of LPSCl and LPSiSOCl were conducted in the force-spectroscopy (FS) scanning mode of an atomic force microscope (AFM, NX-10, Park Systems Co., Ltd.) equipped with a logarithm amplifier module and a silicon probe (PPP-NCHR). During the FS measurements, the cantilever repeatedly approached and retracted from each scanning point (256 × 256 pixels), and the relationship between the vertical (z) displacement and the applied force was recorded. AFM nanoindentation measurements were performed using an atomic force microscope (AFM; NX-10, Park Systems) equipped with a single-crystal diamond tip. The loading force was increased up to 70 μN (7.5 V), while the loading rate was maintained under the instrument's preset conditions. H_2S evolution from the solid electrolyte pellets (100 mg, 10 mm diameter) was monitored for 1 h under humid air conditions (50% RH) in a sealed 2 L chamber using an infrared H_2S detector. The humid atmosphere was generated by mixing dry and water-bubbled air. A 3D X-ray microscope (XRM; SKYSCAN 2214 CMOS Edition, Bruker) equipped with a CMOS2 detector operated in high-aspect-ratio imaging mode was used. The voxel size was set to 0.3 μm, and the X-ray source was operated at 70 kV and 110 μA. Tomographic scans were acquired over 360° with an angular step of 0.2° using three-frame averaging at each projection to improve the signal-to-noise ratio. The total scan time for each sample was approximately 16 h. The acquired projection data were reconstructed using NRecon (Bruker), followed by 3D visualization and analysis using Avizo (Thermo Fisher Scientific).

4.3 | Electrochemical Measurements

Composite cathodes were prepared in a dry room at −50°C under controlled humidity. The cathode active material was $LiNi_{0.7}Co_{0.15}Mn_{0.15}O_2$ (NCM), the solid electrolytes were LPSCl and LPSiSOCl, and the conductive additive was Super-P. The composite cathode materials were homogenized using a vortex mixer. The cathode composition was fixed at a weight ratio of 80:19:1 for the cathode active material (NCM), the solid electrolytes (LPSCl and LPSiSOCl), and Super-P, respectively. Cathode half-cells for ASSBs were fabricated in the same dry room. The cell configuration was Li (-In)|SEs|cathodes. A total of 90 mg of solid electrolyte powders (LPSCl and LPSiSOCl) was first loaded into a polyetheretherketone (PEEK) pressure cell and pressed at 312 MPa, after which the cathode material was loaded. Each cathode (11.4 mg cm^{-2}) with an Al current collector disk (10 mm diameter, 15 μm thickness) was positioned on one face of the solid electrolyte pellet (10 mm diameter). After placing the cathode in the cell, pressure of 437 MPa was applied using a cold press to ensure intimate contact between the cathode and the solid electrolyte layers. A Li-In alloy anode with a Ni current collector disk (10 mm diameter, 10 μm thickness) was then added. The assembled cell was finally compressed to 6.5 N·m (corresponding to 65 MPa) using a torque wrench. The Li disk used for the alloy was 4 mm in diameter and 200 μm thick, and the In disk was 9 mm in diameter and 100 μm thick.

4.4 | Electrochemical Measurements

Electrochemical impedance spectroscopy (EIS) measurements were recorded using a VSP-300 potentiostat (Bio-Logic). The ionic conductivity of the solid electrolytes was evaluated from AC impedance measurements on Ti|SEs|Ti cells with an excitation amplitude of 15.0 mV over the frequency range of 7.0 MHz−0.1 Hz. For Li-In|SEs|cathode cells, the overall resistance was obtained from impedance spectra collected with an excitation amplitude of 15.0 mV over 7.0 MHz-0.01 Hz. In subsequent evaluations of the overall resistance of the cells, OCV-dependent EIS profiles were recorded in constant-current charging mode.

4.5 | Computational Details

Density functional theory (DFT) calculations were performed to evaluate the impact of Si and O doping on the elastic properties of Li_6PS_5Cl. All DFT calculations were performed using the Vienna Ab Initio Simulation Package (VASP) [68]. The interaction between core and valence electrons was described using the projector augmented-wave (PAW) method, and a plane-wave cutoff energy of 520 eV was applied. The exchange–correlation functional was employed within the Perdew–Burke–Ernzerhof (PBE) form of the generalized gradient approximation (GGA) [69].

The Li_6PS_5Cl structure with the $F\bar{4}3m$ space group was taken from the Materials Project database (mp-985592) [70]. To consider the experimentally observed S^{2-}/Cl^- disordering, the bulk structure with 52 atoms was generated by setting the occupancies of the S^{2-} (4d) and Cl^- (4a) sites to 50% [71], using the Python Materials Genomics (pymatgen) code [72] to enumerate all possible

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configurations. Symmetry-equivalent structures were removed based on rotation, inversion, and reflection operations along the c-axis. All distinct structures were fully relaxed, and the lowest-energy 50% S^{2-}/Cl^{-} configuration was adopted for subsequent calculations.

Based on the selected structures, substitutions of Si for P and O for S were introduced to investigate the effects of central cation and anion substitutions on the bulk modulus. Consistent with our experimental results, a single oxygen atom was substituted at the Wyckoff 16e site of the PS_4^{3-} tetrahedra. The bulk modulus values were obtained by fitting the calculated energy–volume (E–V) data for Li_6PS_5Cl, $Li_6Si_{0.25}P_{0.75}S_5Cl$ (LPSiSCl), $Li_6PS_{4.75}O_{0.25}Cl$ (LPSOCl), and $Li_6Si_{0.25}P_{0.75}S_{4.75}O_{0.25}Cl$ (LPSiSOCl) to the Murnaghan equation of state, defined as:

$$E\ (V) = \frac{B_0 V}{B_0'\left(B_0' - 1\right)}\left[B_0'\left(1 - \frac{V_0}{V}\right) + \left(\frac{V_0}{V}\right)^{B_0'} - 1\right] + E_0$$

In this expression, B_0 is the bulk modulus, B_0' is its pressure derivative, V_0 is the equilibrium volume, and E_0 is the constant obtained from the Murnaghan fitting.

For each composition, the E–V model comprised six uniformly strained volumes with V/V_0 ranging from 0.96 to 1.02 of the relaxed value. For each model, we used a plane-wave cutoff of 700 eV and an electronic self-consistency threshold of 1×10^{-8} eV. K-point sampling employed a Monkhorst–Pack 3 × 3 × 3 mesh, and structural relaxations were performed with fixed lattice parameters and fully relaxed atomic positions.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. Y. Jiang, A. Lai, J. Ma, et al., "Fundamentals of the Cathode-Electrolyte Interface in All Solid-State Lithium Batteries," *ChemSusChem* 16 (2023): 202202156.

2. J. Mun, T. Song, M.-S. Park, et al., "Paving the Way for Next-Generation All Solid-State Batteries Dry Electrode Technology," *Advanced Materials* 37 (2025): 2506123.

3. H. Xu, S. Yang, and B. Li, "Pressure Effects and Countermeasures in Solid-State Batteries: A Comprehensive Review," *Advanced Energy Materials* 14 (2024): 2303539.

4. B. N. Yun, S. Lee, W. D. Jung, et al., "Facile Method for the Formation of Intimate Interfaces in Sulfide Based All-Solid-State Batteries," *ACS Applied Material Interfaces* 14 (2022): 9242.

5. J. Kim, M. J. Kim, J. Kim, et al., "High-Performance All-Solid-State Batteries Enabled By Intimate Interfacial Contact Between the Cathode and Sulfide-Based Solid Electrolytes," *Advanced Functional Materials* 33 (2023): 2211355.

6. A. O. Kondrakov, A. Schmidt, J. Xu, et al., "Anisotropic Lattice Strain and Mechanical Degradation of High- and Low-Nickel NCM Cathode Materials for Li-Ion Batteries," *The Journal of Physical Chemistry C* 121 (2017): 3286.

7. R. A. Ahmed, N. Ebechidi, I. Reisya, et al., "Pressure-Induced Interfacial Contacts and the Deformation in All Solid-State Li-Ion Batteries," *Journal of Power Sources* 521 (2022): 230939.

8. J. Zhang, J. Lv, W. Lu, et al., "$Li_{10}GeP_2S_{12}$-Based Solid Electrolyte Induced by High Pressure for All-Solid-State Batteries: A Facile Strategy of Low-Grain-Boundary-Resistance," *Chemical Engineering Journal* 487 (2024): 150452.

9. Y. Lin, H. Liu, Y. Hu, et al., "Sub-Micrometer Li_6PS_5Cl Regulated Cathodic Li Kinetics in Sulfide Based Sll-Solid-State Batteries," *Physical Chemistry Chemical Physics* 27 (2025): 3455–3462.

10. J. Han, A. Vu, J. J. Kim, et al., "Room-Temperature Fabrication of Garnet-Type Solid-Electrolyte: Optimizing Particle Size for High Ionic Conductivity," *Chemical Engineering Journal* 481 (2024): 148645.

11. S. Farzanian, J. Vazquez Mercado, and I. Shozib, "Mechanical Investigations of Composite Cathode Degradation in All-Solid-State Batteries," *ACS Applied Energy Materials* 6 (2023): 9615.

12. D. Lee and A. Manthiram, "Stable Cycling with Intimate Contacts Enabled by Crystallinity-Controlled PTFE-Based Solvent-Free Cathodes in All-Solid-State Batteries," *Small Methods* 7 (2023): 2201680.

13. J. Oh, S. H. Choi, B. Chang, et al., "Elastic Binder for High-Performance Sulfide-Based All-Solid-State Batteries," *ACS Energy Letters* 7 (2022): 1374.

14. Y. Sun, C. Gui, Y. Zhang, et al., "An Adhesive Interface between Hydrogel Electrolyte and Electrode for Low-Temperature Solid-State Capacitive Devices," *Journal of Energy Storage* 102 (2024): 114061.

15. D. L. R. Melvin, M. Siniscalchi, D. Spencer-Jolly, et al., "High Plating Currents without Dendrites at the Interface between a Lithium Anode and Solid Electrolyte," *Nature Energy* 10 (2025): 1205.

16. O. Korjus, S. Mitra, Q. Berrod, et al., "Investigating the Densification of Li_6PS_5Cl Solid Electrolyte through Multi-Scale Characterization Techniques," *Energy Storage Materials* 82 (2025): 104589.

17. I. A. Sokolov, A. A. Il'in, Y. P. Tarlakov, et al., "Structure and Physicochemical Properties of Glasses in the Li_2S–$LiPO_3$ System," *Glass Physics and Chemistry* 29 (2003): 282.

18. B. Hu, S. Zhang, Z. Ning, et al., "Deflecting Lithium Dendritic Cracks in Multi-Layered Solid Electrolytes," *Joule* 8 (2024): 2623.

19. A. Sakuda, A. Hayashi, and M. Tatsumisago, "Sulfide Solid Electrolyte with Favorable Mechanical Property for All-Solid-State Lithium Battery," *Science Reports* 3 (2013): 2261.

20. J. Yun, H. R. Shin, T. D. Hoang, et al., "Deciphering the Critical Degradation Factors of Solid Composite Electrodes with Halide Electrolytes: Interfacial Reaction Versus Ionic Transport," *Energy Storage Materials* 59 (2023): 102787.

21. M. Palmer, L. Qian, V. K. Singh, et al., "Contact Mechanics and Electrochemical Properties of the $Li_{6.5}La_3Zr_{1.5}Ta_{0.5}O_{12}$| $Li_{2.25}Zr_{0.75}Fe_{0.25}Cl_6$ Hetero-Electrolyte Interface in a Low-Pressure Regime," *Solid State Ionics* 428 (2025): 116948.

22. M. Balaish, K. J. Kim, H. Chu, et al., "Emerging Processing Guidelines for Solid Electrolytes in the Era of Xxide-Based Solid-State Batteries," *Chemical Society Reviews* 54 (2025): 8925.

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23. S. Hong, K. H. Shin, S. Kim, et al., "Mechanical-Stress-Induced Lithiation and StructuralEvolution Driven by Excess Lithium Predisposing Short Circuits at the Surface of Garnet Solid Electrolytes," *Advanced Energy Materials* 15 (2024): 2402666.

24. Y. Ren, T. Danner, A. Moy, et al., "Oxide-Based Solid-State Batteries: A Perspective on Composite Cathode Architecture," *Advanced Energy Materials* 13 (2023): 2201939.

25. J. Gu, Z. Liang, J. Shi, et al., "Electrochemo-Mechanical Stresses and Their Measurements in Sulfide-Based All-Solid-State Batteries: A Review," *Advanced Energy Materials* 13 (2023): 2203153.

26. R. Koerver, W. Zhang, L. de Biasi, et al., "Chemo-Mechanical Expansion of Lithium Electrode Materials – on the Route Tomechanically Optimized All-Solid-State Batteries," *Energy & Environmental Science* 11 (2018): 2142.

27. J. Zhang, L. Li, C. Zheng, et al., "Silicon-Doped Argyrodite Solid Electrolyte Li_6PS_5I with Improved Ionic Conductivity and Interfacial Compatibility for High-Performance All-Solid-State Lithium Batteries," *ACS Applied Material Interfaces* 12 (2020): 41538.

28. W. D. Jung, M. Jeon, S. S. Shin, et al., "Functionalized Sulfide Solid Electrolyte with Air-Stable and Chemical-Resistant Oxysulfide Nanolayer for All-Solid-State Batteries," *ACS Omega* 5 (2020): 26015.

29. S. J. Park, Y. Subramanian, Y. J. Jung, et al., "Enhanced Lithium Dendrite Suppression Ability through SiO_2 Substitution in Superionic Halogen-Rich Argyrodites and Their Application in All-Solid-State Lithium Batteries," *Journal of Industrial and Engineering Chemistry* 135 (2024): 416.

30. S. Wang, Y. Zhang, X. Zhang, et al., "High-Conductivity Argyrodite Li_6PS_5Cl Solid Electrolytes Prepared via Optimized Sintering Processes for All-Solid-State Lithium−Sulfur Batteries," *ACS Applied Material Interfaces* 10 (2018): 42279.

31. H. Jo, B. Kim, B.-M. Ahn, et al., "High-Purity Li_6PS_5Cl Sulfide Solid Electrolytes via Liquid-Phase Method Synthesis for High-Performance All-Solid-State Li-Ion Batteries: Particle-Scale Analysis of Formation with Precursor and Solvent Optimization," *Chemical Engineering Journal* 522 (2025): 166951.

32. L. Ming, Q. Luo, C. Wei, et al., "Unravelling th O-doping Effect on Chemical/Electrochemical Stability of $Li_{5.5}PS_{4.5}Cl_{1.5}$ for All-Solid-State Lithium Batteries," *Next Materials* 5 (2024): 100233.

33. Q. Luo, C. Liu, L. Li, et al., "O-Doping Strategy Enabling Enhanced Chemical/Electrochemical Stability of Li_3InCl_6 for Superior Solid-State Battery Performance," *Journal of Energy Chemistry* 99 (2024): 484.

34. J. Yang, Y. Wang, Z. Jiang, et al., "Lanthanum and Oxygen Co-Doping of Lithium Argyrodite for Enhanced Humid Air Stability and Lithium Metal Compatibility," *Electrochimica Acta* 535 (2025): 146619.

35. S. Ohno, B. Helm, T. Fuchs, et al., "Further Evidence for Energy Landscape Flattening in the Superionic Argyrodites $Li_{6+x}P_{1-x}M_xS_5I$ (M= Si, Ge, Sn)," *Chemistry of Materials* 31 (2019): 4936.

36. L. Zhou, A. Assoud, Q. Zhang, et al., "New Family of Argyrodite Thioantimonate Lithium Superionic Conductors," *Journal of the American Chemical Society* 141 (2019): 19002.

37. J. Auvergniot, A. Cassel, J.-B. Ledeuil, et al., "Interface Stability of Argyrodite Li_6PS_5Cl toward $LiCoO_2$, $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$, and $LiMn_2O_4$ in Bulk All-Solid-State Batteries," *Chemistry of Materials* 29 (2017): 3883.

38. K. H. Park, Q. Bai, D. H. Kim, et al., "Design Strategies, Practical Considerations, and New Solution Processes of Sulfide Solid Electrolytes for All-Solid-State Batteries," *Advanced Energy Materials* 8 (2018): 1800035.

39. C. Dietrich, R. Koerver, M. W. Gaultois, et al., "Spectroscopic Characterization of Lithium Thiophosphates by XPS and XAS – a Model to Help Monitor Interfacial Reactions in All-Solid-State Batteries," *Physical Chemistry Chemical Physics* 20 (2018): 20088.

40. Z. Sun, Y. Lai, N. Lv, et al., "Insights on the Properties of the O-Doped Argyrodite Sulfide Solid Electrolytes ($Li_6PS_{5-x}ClO_x$, x=0−1-)," *ACS Applied Material Interfaces* 13 (2021): 54924.

41. H. Shi, T. Xu, D. Wang, et al., "Oxygen Substitution at the Unbonded S Site for Excellent Wet-Air Stability and Lithium Compatibility of Br-Rich Li-Argyrodite Solid-State Electrolytes," *Journal of Materials Chemistry A* 12 (2024): 29009.

42. J. Liang, X. Li, Y. Zhao, et al., "An Air-Stable and Dendrite-Free Li Anode for Highly Stable All-Solid-State Sulfide-Based Li Batteries," *Advanced Energy Materials* 9 (2019): 1902125.

43. M. Yoshimizu, R. Kobayashi, M. Saegusa, et al., "Photocatalytic Hydrogen Evolution over β-Iron Silicide under Infrared-Light Irradiation," *Chemical Communications* 51 (2015): 2818.

44. A. Raza, J. Y. Jung, C. H. Lee, et al., "Swelling-Controlled Double-Layered $SiO_x/Mg_2SiO_4/SiO_x$ Composite with Enhanced Initial Coulombic Efficiency for Lithium-Ion Battery," *ACS Applied Material Interfaces* 13 (2021): 7161.

45. Y. Park, Y. Shim, J. Lee, et al., "Impacts of Site-Selective Oxygen Introduction on Structural Stabilization, Moisture Stability, and Battery Performance in Sulfide-Based Argyrodite," *Energy Storage Materials* 75 (2025): 104078.

46. P. Ghorbanzade, A. Pesce, M. Armand, et al., "Unveiling the Reactivity and th Li-Ion Exchange at the PEO-Li_6PS_5Cl Interphase: Insights from Solid-State NMR," *Small Structure* 5 (2024): 2400139.

47. P. Wang, H. Liu, S. Patel, et al., "Fast Ion Conduction and Its Origin in $Li_{6-x}PS_{5-x}Br_{1+x}$," *Chemistry of Materials* 32 (2020): 3833.

48. A. Kato, M. Nose, M. Yamamoto, et al., "Mechanical Properties of Sulfide Glasses in All-Solid-State Batteries," *Journal of the Ceramic Society of Japan* 126 (2018): 719.

49. H. Zhou, C. Persson, W. Xia, et al., "Effects of N/Si Ratio on Mechanical Properties of Amorphous Silicon Nitride Coating," *Materials Research Express* 10 (2023): 115403.

50. A. Kato, M. Nagao, A. Sakuda, et al., "Evaluation of Young's Modulus of Li_2S–P_2S_5–P_2O_5 Oxysulfide Glass Solid Electrolytes," *Journal of the Ceramic Society of Japan* 122 (2014): 552–555.

51. K. Hikima, M. Totani, S. Okokata, et al., "Mechanical Properties of Sulfide-Type Solid Electrolytes Analyzed by Indentation Methods," *ACS Applied Energy Materials* 5 (2022): 2349.

52. H. Kim, K. Hikima, K. Watanabe, et al., "Mechanical Properties of $Li_{10.35}Ge_{1.35}P_{1.65}S_{12}$ with Different Particle Sizes," *Materials Transactions* 65 (2024): 861–866.

53. F. Wu, Q. Li, L. Chen, et al., "Use of Ce to Reinforce the Interface of Ni-Rich $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ Cathode Materials for Lithium-Ion Batteries under High Operating Voltage," *ChemSusChem* 12 (2019): 935.

54. H.-H. Ryu, B. Namkoong, J.-H. Kim, et al., "Capacity Fading Mechanisms in Ni-Rich Single-Crystal NCM Cathodes," *ACS Energy Letters* 6 (2021): 2726.

55. Y. Zhao, Y. Bao, W. Zhang, M. Guan, and Y. Zhuang, "The Effect on Electrochemical Properties of Ti^{4+} and V^{5+} Doping in $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$," *Journal of Alloys and Compounds* 958 (2023): 170451.

56. P. Lu, Y. Wu, D. Wu, et al., "Rate-Limiting Mechanism of All-Solid-State Battery Unravelled by Low-Temperature Test-Analysis Flow," *Energy Storage Materials* 67 (2024): 103316.

57. Z. Tang, A. Morchhale, J. R. Sayre, and J. H. Kim, "In-Situ Electrochemical Characterization of Dynamic Void Formation and Lithium Dendrite in All-Solid-State Batteries," *Journal of Energy Storage* 107 (2025): 114966.

58. R. He, Y. He, W. Xie, B. Guo, and S. Yang, "Comparative Analysis for Commercial Li-Ion Batteries Degradation Using the Distribution of Relaxation Time Method Based on Electrochemical Impedance Spectroscopy," *Energy* 263 (2023): 125972.

59. C.-Y. Yu, J. Choi, J. Dunham, et al., "Time-Resolved Impedance Spectroscopy Analysis of Aging in Sulfide-Based All-Solid-State Battery Full-Cells Using Distribution of Relaxation Times Technique," *Journal of Power Sources* 597 (2024): 234116.

16163028, 0, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.78249 by Chung-Ang University, Wiley Online Library on [27/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

60. D. H. S. Tan, E. A. Wu, H. Nguyen, et al., "Elucidating Reversible Electrochemical Redox of Li_6PS_5Cl Solid Electrolyte," *ACS Energy Letters* 4 (2019): 2418–2427.

61. T. K. Schwietert, V. A. Arszelewska, and C. Wang, "Clarifying the Relationship Between Redox Activity and Electrochemical Stability in Solid Electrolytes," *Nature Materials* 19 (2020): 428–435.

62. A. Banerjee, X. Wang, C. Fang, E. A. Wu, and Y. S. Meng, "Interfaces and Interphases in All-Solid-State Batteries with Inorganic Solid Electrolytes," *Chemical Reviews* 120 (2020): 6878–6933.

63. S. Y. Kim, H. Cha, R. Kostecki, and G. Chen, "Composite Cathode Design for High-Energy All-Solid-State Lithium Batteries with Long Cycle Life," *ACS Energy Letters* 8 (2023): 521–528.

64. D. Lee, Y. Shim, E. Choi, et al., "Alleviating Kinetical Delamination Induced by Localized Cathode Contact via Electrochemo-Mechanical Modeling in All-Solid-State Batteries," *Joule* 9 (2025): 102046.

65. X. Li, L. Jin, D. Song, et al., "$LiNbO_3$-Coated $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ Cathode with High Discharge Capacity and Rate Performance for All-Solid-State Lithium Battery," *Journal of Energy Chemistry* 40 (2020): 39–45.

66. F. Walther, F. Strauss, X. Wu, et al., "The Working Principle of a $Li_2CO_3/LiNbO_3$ Coating on NCM for Thiophosphate-Based All-Solid-State Batteries," *Chemistry of Materials* 33 (2021): 2110–2125.

67. K. Watanabe, H. Nakayama, H.-S. Kim, et al., "Sn vs. Ge: Effects of Elastic and Plastic Deformation of LGPS-type Solid Electrolytes on Charge-Discharge Properties of Composite Cathodes for All-solid-state Batteries," *Electrochemistry* 93 (2025): 063019.

68. G. Kresse and J. Furthmüller, "Efficient Iterative Schemes for Ab Initio Total-Energy Calculations Using a Plane-Wave Basis Set," *Physical Review B* 54 (1996): 11169.

69. J. P. Perdew, K. Burke, and M. Ernzerhof, "Generalized Gradient Approximation Made Simple," *Physical Review Letters* 77 (1996): 3865.

70. A. Jain, S. P. Ong, G. Hautier, et al., "Commentary: The Materials Project: A Materials Genome Approach to Accelerating Materials Innovation," *APL Materials* 1 (2013): 011002.

71. M. A. Kraft, S. P. Culver, M. Calderon, et al., "Influence of Lattice Polarizability on the Ionic Conductivity in the Lithium Superionic Argyrodites Li_6PS_5X (X = Cl, Br, I)," *Journal of the American Chemical Society* 139 (2017): 10909–10918.

72. S. P. Ong, W. D. Richards, A. Jain, et al., "Python Materials Genomics (pymatgen): A Robust, Open-Source Python Library for Materials Analysis," *Computational Materials Science* 68 (2013): 314–319.

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